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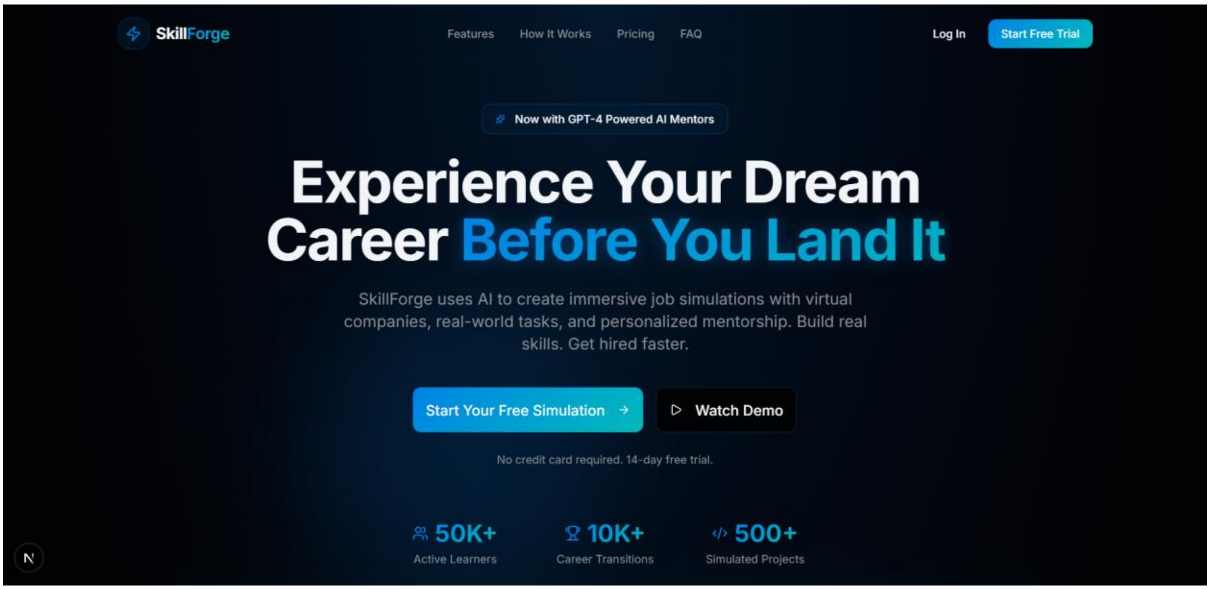
Project Title

SkillForge – AI Powered Career Simulation Platform with DevOps Deployment

1.Introduction

SkillForge is an AI-powered career simulation and learning platform designed to help students gain practical industry experience before entering the professional world. The platform provides coding tasks, mentorship, interview preparation, and virtual company simulations.

This project was deployed using DevOps technologies including GitHub, GitHub Actions, Docker, AWS EC2, and Elastic IP.



2. Objectives

- To provide practical industry-oriented learning.
- To deploy a web application using DevOps tools.
- To implement CI/CD workflow using GitHub Actions.
- To host the project publicly using AWS EC2.

3. Technologies Used

Frontend

- Next.js
- React.js
- Tailwind CSS

Backend Runtime

- Node.js

DevOps & Deployment

- GitHub
- GitHub Actions
- Docker
- AWS EC2
- Ubuntu Linux
- Elastic IP

4)CI/CD Pipeline Using GitHub Actions

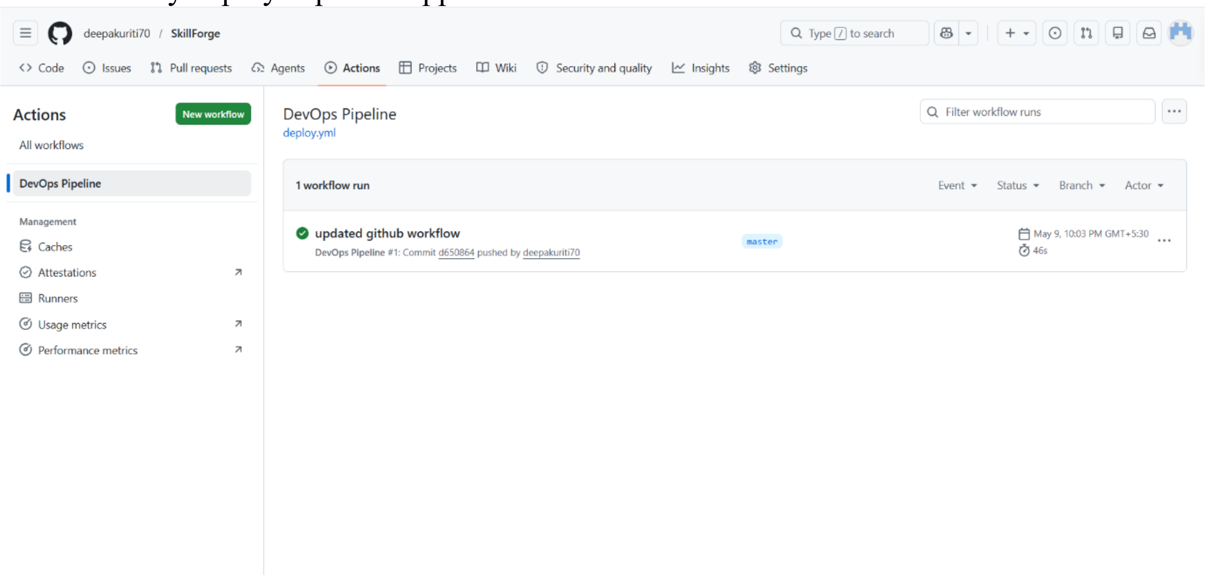
GitHub Actions was used to implement CI/CD pipeline.

Whenever code is pushed to GitHub repository, the workflow automatically triggers.

CI/CD Workflow

Developer Code
↓
GitHub Repository
↓
GitHub Actions Workflow
↓
Docker Build
↓
AWS EC2 Deployment
↓
Public Website

CI – Continuous Integration
Automatically integrates and checks code changes.
CD – Continuous Deployment
Automatically deploys updated application.



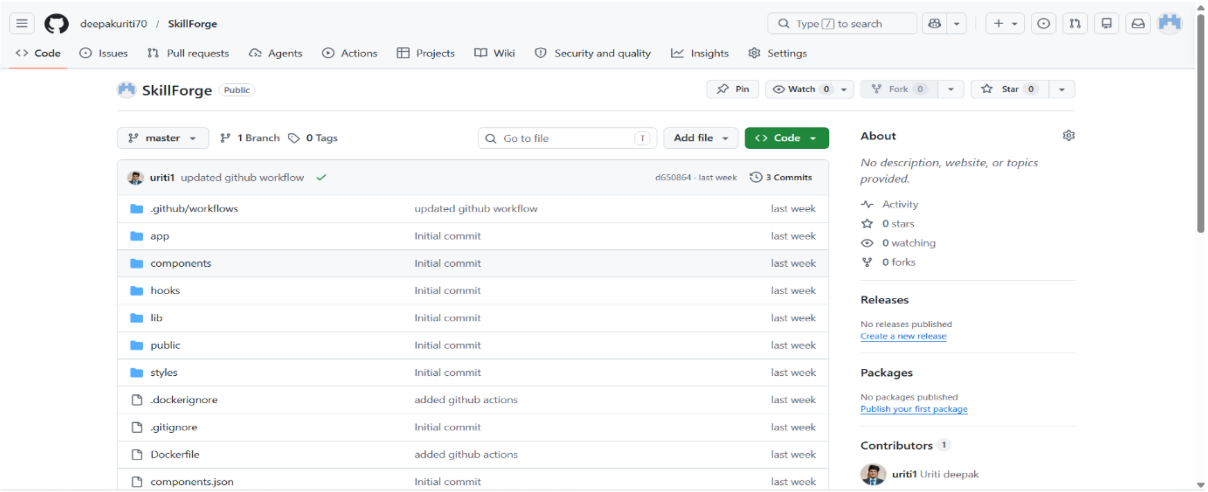
Steps to open my Project:

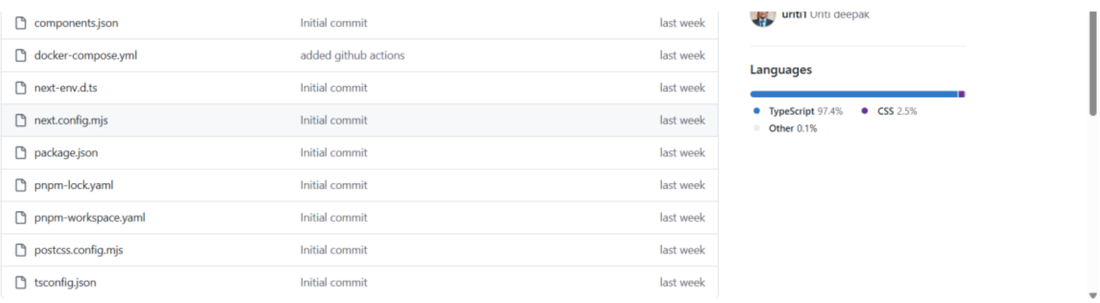
1) GitHub Repository

Here, I stored my complete project source code in GitHub for version control and collaboration.

GitHub also helps in maintaining project versions and tracking code changes.

- Repository contains complete project files.
- Git used for version control.





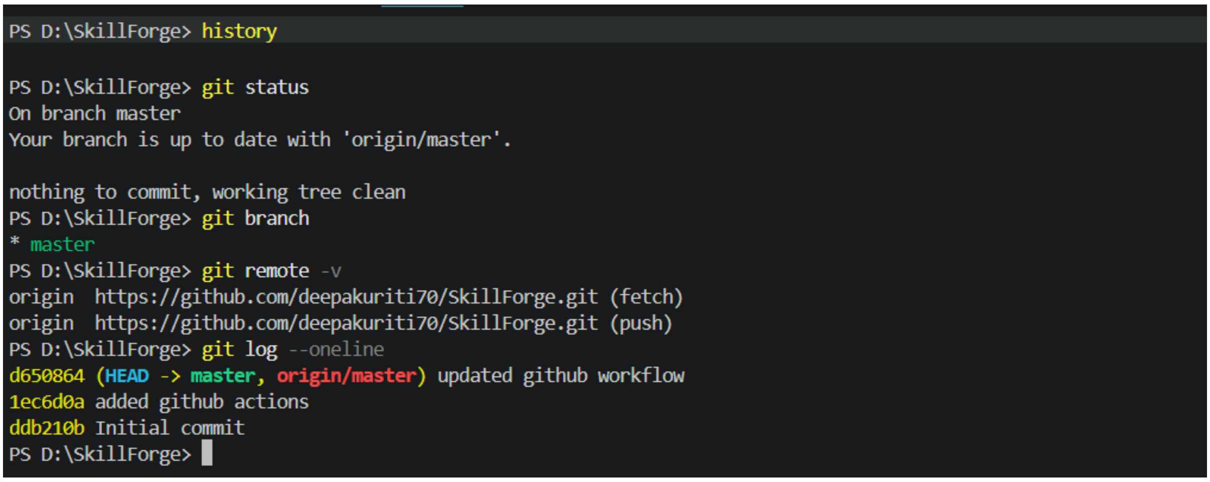
2)Git Commands Used:

git clone https://github.com/deepakuriti70/SkillForge.git

git add .

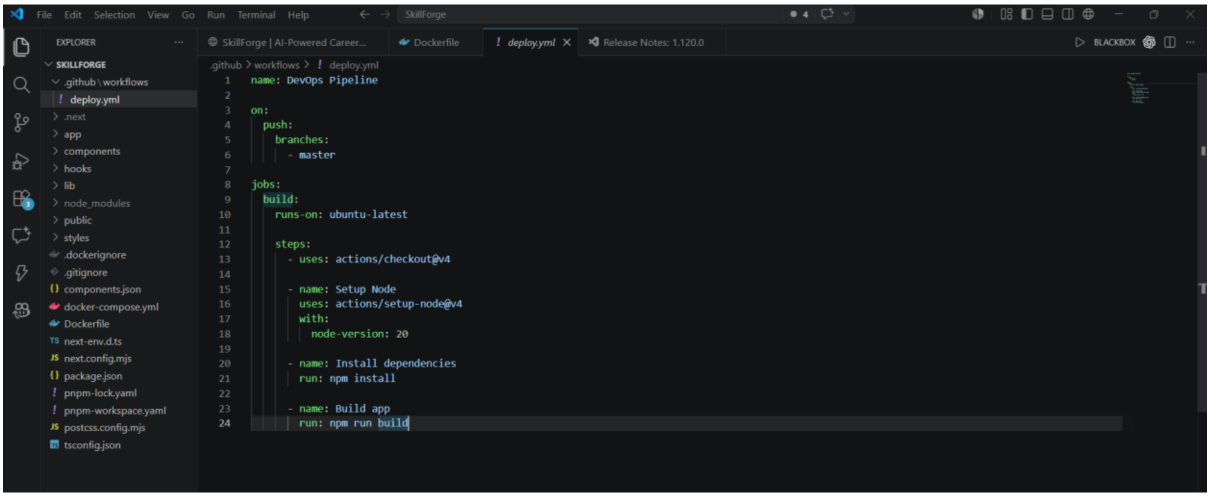
git commit -m "updated workflow"

git push origin master

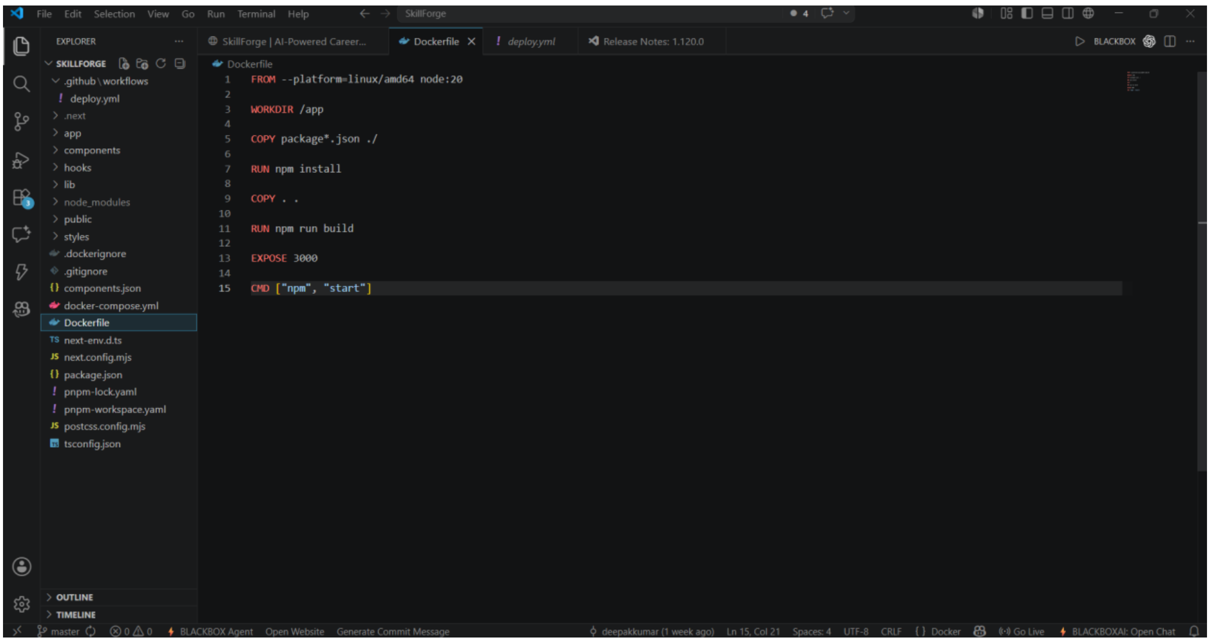


3) Project Structure in VS Code

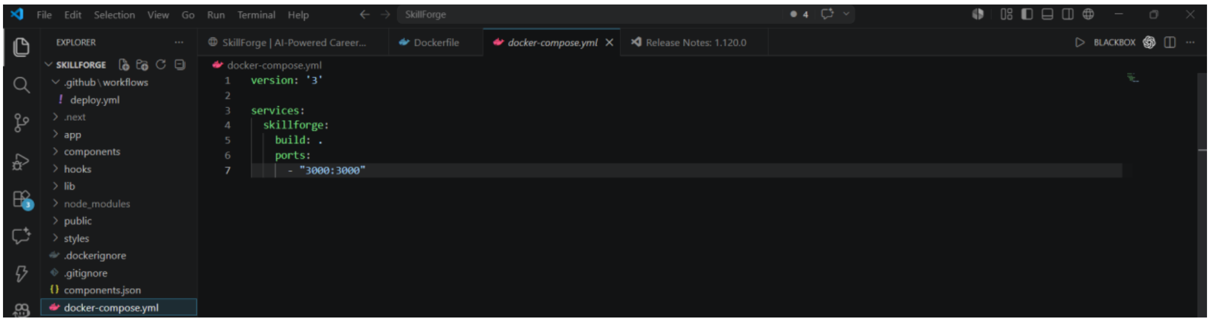
The project was developed using Visual Studio Code



This folder contains GitHub Actions CI/CD workflow configuration.



Dockerfile contains instructions to build Docker image and run the application inside contain.



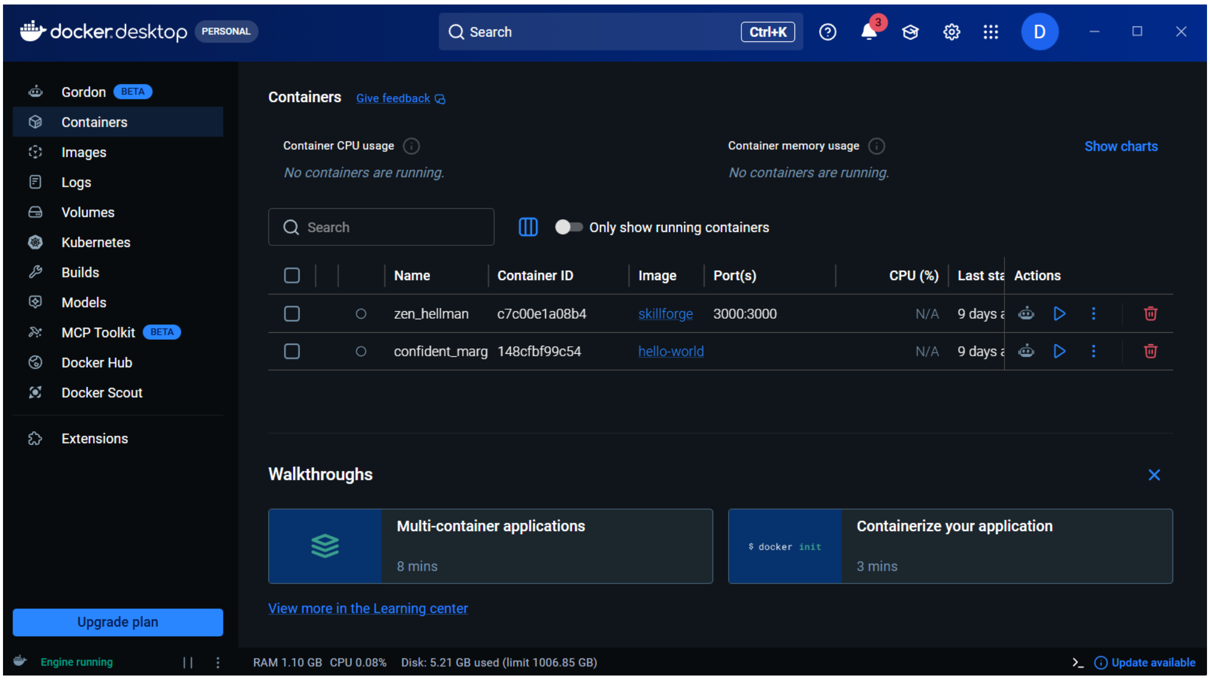
Dockerfile contains instructions to build Docker image and run the application inside contain.

Used to manage multi-container Docker applications.

3) Docker Containerization

Docker was used to containerize the SkillForge application.

Containerization helps the application run consistently in all environments.



Docker Desktop was used to manage the SkillForge container and monitor Docker images and ports during deployment.

```
ubuntu@ip-172-31-40-192:~$ cd ~/SkillForge
ubuntu@ip-172-31-40-192:~/SkillForge$ sudo docker build -t skillforge .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  773.1kB
Step 1/8 : FROM --platform=linux/amd64 node:20
 ---> 8f693eaa7e0a
Step 2/8 : WORKDIR /app
 ---> Using cache
 ---> 948fa8647ae7
Step 3/8 : COPY package*.json ./
 ---> Using cache
 ---> 22ab480d9386
Step 4/8 : RUN npm install
 ---> Using cache
 ---> f336aaab1dde
Step 5/8 : COPY . .
 ---> Using cache
 ---> 2c062b036def
Step 6/8 : RUN npm run build
 ---> Using cache
 ---> f3eee5cb8a93
Step 7/8 : EXPOSE 3000
 ---> Using cache
 ---> 435ffa26a350
Step 8/8 : CMD ["npm", "start"]
 ---> Using cache
 ---> ec9a8e1ac1c6
Successfully built ec9a8e1ac1c6
Successfully tagged skillforge:latest
ubuntu@ip-172-31-40-192:~/SkillForge$ `
```

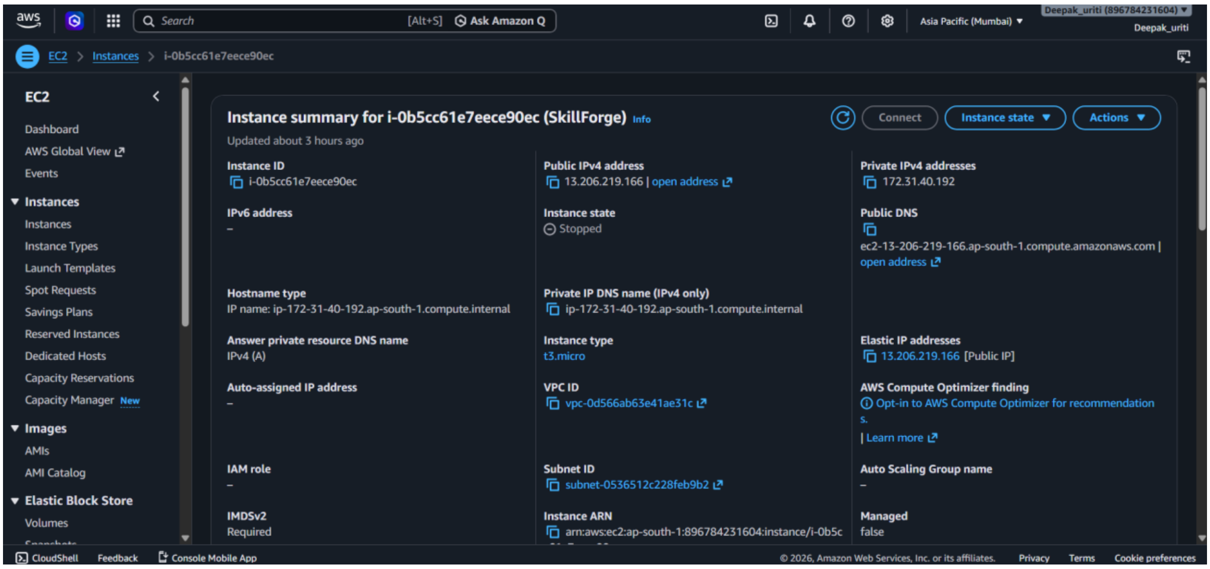
Docker image for the SkillForge application was successfully built using the Dockerfile and tagged as skillforge:latest for deployment.

Dockerfile contains instructions to build Docker image and run the application inside contain.

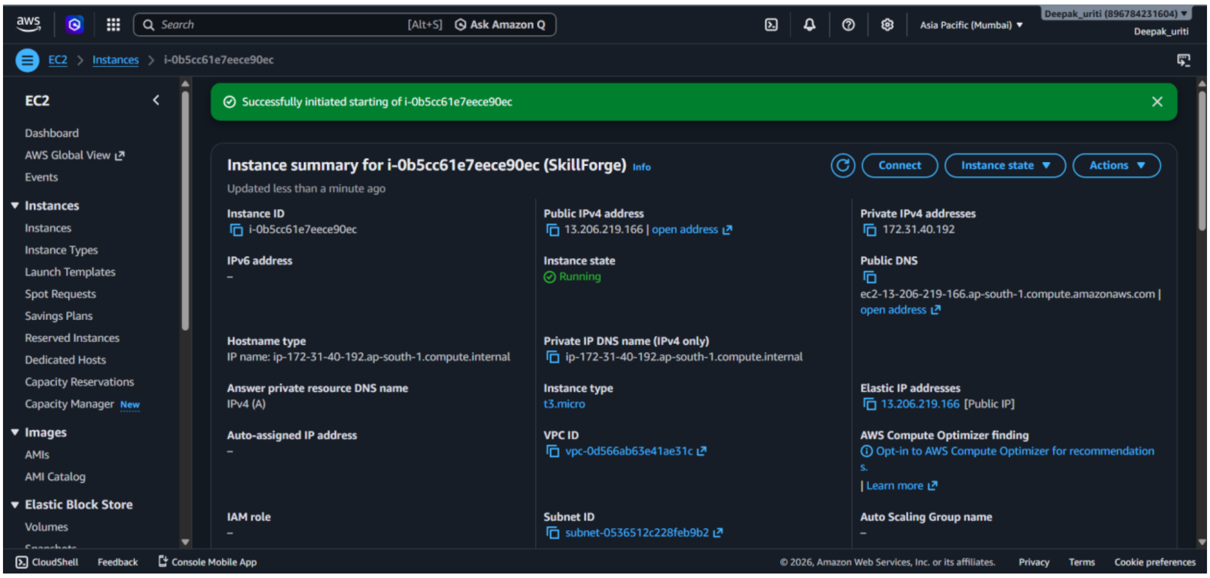
Used to manage multi-container Docker applications.

4)After turn on the Docker we have to open the AWS console and turn on the Instance.

Open AWS console -> EC2 -> Instances -> SkillForge -> Instance state -> Start.



In this my instance is Stopped.



In this the instance is started and working.

5)After this open the Powershell

1. Connect to Server

ssh -i D:\skillforge-key.pem ubuntu@13.206.219.166

2. Check Containers

sudo docker ps -a

3. Start Your Container

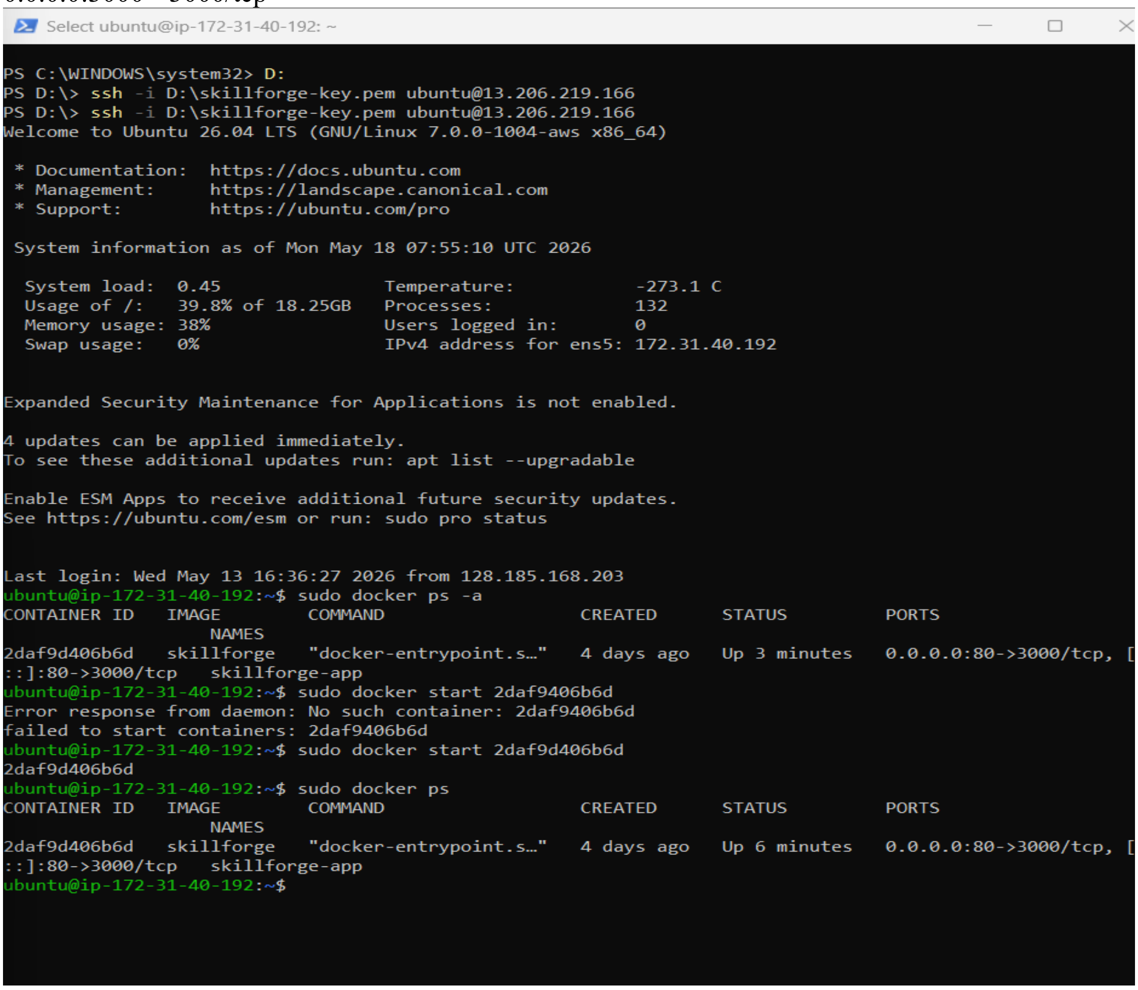
Copy the CONTAINER ID and run:
sudo docker start CONTAINER_ID

sudo docker start 9988f0b7f3f4

3. Verify Running

sudo docker ps

You should see:
0.0.0.0:3000->3000/tcp



5. Open Browser

http://13.206.219.166/

6)The SkillForge application was successfully deployed publicly using AWS EC2 and Docker.

Public URL

http://13.206.219.166/